

Geography Ch-16 (Advanced): The Indian Monsoon Mechanism - Onset, Branches, Break & Withdrawal

Tier 2: Advanced | Source: D.R. Khullar + Majid Husain (Geography of India)
India-applied grounded in Qdrant upsc_static book DB + live Current Affairs anchor

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

The Indian Monsoon: Traditional & Modern Theories, Two Branches, 'Break' & Retreat

GS-I (Physical Geography of India) Geography

■ NEWS TRIGGER

Advanced application (D.R. Khullar; Majid Husain, Geography of India). The Indian monsoon is explained by classical thermal theory (Halley) and the modern dynamic theory (jet streams, Tibetan Plateau, ITCZ, ENSO/IOD, Somali jet, Tropical Easterly Jet).

■ INTRO & ORIGIN

Beyond the simple land-sea contrast, the modern theory of the Indian monsoon integrates the northward shift of the ITCZ, the Tibetan Plateau as a summer heat engine, the withdrawal of the Subtropical Westerly Jet north of the Himalayas, the onset of the Tropical Easterly Jet, and the Somali (Findlater) low-level jet - all timed to produce the June 'burst'.

Halley's thermal theory (differential heating) is now supplemented by the dynamic/jet-stream theory. The monsoon splits into two branches on hitting the Indian peninsula.

■ KEY DATA

Element	Role	Effect
Arabian Sea branch	Strikes Western Ghats	Heavy windward rain; rain-shadow Deccan
Bay of Bengal branch	Enters Ganga plain & NE	Rain up the plains; Meghalaya wettest
Tibetan Plateau	Elevated summer heat source	Drives upper-air low pressure, TEJ
Subtropical Westerly Jet	Withdraws N of Himalayas	Pre-condition for onset
Somali (Findlater) Jet	Cross-equatorial low-level jet	Feeds SW monsoon moisture

■ STATIC THEORY

- **Two branches (Khullar):** the Arabian Sea branch strikes the Western Ghats (heavy windward rain, dry leeward Deccan rain-shadow) and reaches Mumbai/Gujarat; the Bay of Bengal branch sweeps up the Ganga plains and is deflected by the Himalayas towards the NE (Meghalaya - world's wettest).
- **Modern dynamic factors:** northward shift of the ITCZ; the Tibetan Plateau as an elevated heat source; withdrawal of the Subtropical Westerly Jet and establishment of the Tropical Easterly Jet (TEJ) over peninsular India; the cross-equatorial Somali/Findlater low-level jet transporting moisture.
- **'Break' in the monsoon:** spells of dry weather during the rainy season when the monsoon trough (ITCZ) shifts (e.g. towards the Himalayan foothills, giving heavy foothill rain but dry plains) - critical for crop stress.
- **ENSO & IOD:** El Nino weakens the monsoon; positive IOD strengthens it - the modern teleconnection additions to monsoon theory.
- **Retreat / NE monsoon:** from October the monsoon withdraws; reversed NE winds pick up Bay-of-Bengal moisture and give the Tamil Nadu/Coromandel coast its main (winter) rain - the 'retreating' or NE monsoon.

■ MUST-KNOW FACTS

1. Two branches: Arabian Sea (Western Ghats) + Bay of Bengal (Ganga plain & NE).
2. Tibetan Plateau = elevated summer heat source; drives the Tropical Easterly Jet (TEJ).

■ UPSC TRAPS

- ✗ Tamil Nadu gets most of its rain from the SW monsoon.
- ✓ Coastal Tamil Nadu (Coromandel) gets its main rain from the RETREATING/NE monsoon (Oct-Dec), being in the SW-monsoon rain-shadow.

3. Somali/Findlater Jet = cross-equatorial low-level jet feeding SW-monsoon moisture.
4. 'Break monsoon' = dry spells when the monsoon trough (ITCZ) shifts position.
5. Retreating (NE) monsoon gives Tamil Nadu its chief rainfall in Oct-Dec.

✗ The Tibetan Plateau cools India and weakens the monsoon.

✓ In summer the Plateau is an elevated HEAT source, strengthening the upper-air circulation (TEJ) that drives the monsoon.

✗ A 'break' in the monsoon means the monsoon has withdrawn.

✓ A 'break' is a temporary dry spell WITHIN the rainy season due to a shift of the monsoon trough, not the seasonal withdrawal.

■ MAINS ANGLE

Critically compare the classical thermal and modern dynamic theories of the Indian monsoon, and explain how 'break' periods and the retreating monsoon shape India's agricultural geography.

■ STUDY LINK

Geography -> Indian Monsoon Mechanism -> branches, jets, break, retreat; connects to jet streams (Ch-13 Advanced)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA Anchor: 'Mission Mausam' - Modernising India's Monsoon & Weather Forecasting

GS-III (Science & Tech / Disaster Management) + GS-I (Geography)
Geography - Current Affairs

■ NEWS TRIGGER

Mission Mausam - launched by the Ministry of Earth Sciences (2024) with an outlay of about Rs 2,000 crore (approx.) over two years - aims to make India 'weather-ready and climate-smart' through denser Doppler radars, wind profilers, AI/ML nowcasting and eventually experimental cloud-seeding, coinciding with IMD's 150th year.

■ INTRO & ORIGIN

Predicting the monsoon accurately is a national priority given its economic weight. Mission Mausam is the flagship push to upgrade India's observation and forecasting backbone - directly improving how the monsoon's onset, 'break' and withdrawal are predicted.

Ministry of Earth Sciences (MoES) initiative announced in 2024; builds on IMD/IITM/NCMRWF capacity. (NOTE: exact outlay reported around Rs 2,000 crore - treat figure as approximate.)

■ KEY DATA

Pillar	Tool	Purpose
Denser observation	Doppler radars, wind profilers, AWS	Better real-time monsoon tracking
Modelling	AI/ML + higher-resolution models	Sharper nowcasts & seasonal forecasts
Weather modification	Experimental cloud seeding	Explore managed rainfall/hail suppression
Delivery	Urban & agro-met advisories	Farmer + disaster preparedness

■ STATIC THEORY

• **Mission Mausam objectives:** improve forecast accuracy and lead time from nowcasting to long-range; expand Doppler Weather Radar coverage; strengthen satellite meteorology (INSAT series with ISRO); AI/ML-based prediction; city-specific and agro-meteorological services.

- **Why it matters for the monsoon:** accurate onset/withdrawal and 'break' prediction directly aids sowing decisions, reservoir management and cyclone/flood warnings.
- **Institutional web:** MoES houses IMD, IITM (Pune), INCOIS and NCMRWF - the forecasting ecosystem Mission Mausam upgrades.

■ MUST-KNOW FACTS

1. Mission Mausam launched by the Ministry of Earth Sciences in 2024; outlay ~Rs 2,000 crore (approx., over ~2 years).
2. Focus: Doppler radars, wind profilers, AI/ML nowcasting, and experimental cloud seeding.
3. Coincides with IMD's 150th anniversary (IMD founded 1875).
4. MoES ecosystem: IMD, IITM Pune, INCOIS, NCMRWF.
5. Goal: make India 'weather-ready and climate-smart' for agriculture and disaster management.

■ UPSC TRAPS

- ✗ Mission Mausam is the same as the Ministry of Culture's 'Project Mausam'.
- ✓ Different: 'Project Mausam' is a culture/maritime-heritage initiative; 'Mission Mausam' is the MoES weather-forecasting modernisation programme.
- ✗ Mission Mausam is run by ISRO.
- ✓ It is a Ministry of Earth Sciences (MoES) initiative; ISRO is a satellite-data partner, not the lead.

■ MAINS ANGLE

Discuss how Mission Mausam strengthens India's capacity to forecast the monsoon and extreme weather, and its significance for climate-resilient agriculture and disaster management.

■ STUDY LINK

Geography -> Indian Monsoon forecasting + GS-III S&T/DM; connect to IMD, MoES and monsoon prediction